Total Mark: **15** Duration: **40 minutes**

**THEORY PART**

I. **MULTIPLE CHOICE**. (10 x 1 = 10 marks)
Encircle the letter of the correct answer that corresponds to the given statement.

1. Which of the following best defines the Internet of Things (IoT)?
   A. A network of computers connected for sharing files.
   B. A network of physical objects embedded with sensors and software to exchange data.
   C. A collection of internet-based applications for communication.
   D. A high-speed wireless network connecting personal devices.

2. What role do actuators play in an IoT system?
   A. They collect data from sensors.
   B. They process data from controllers.
   C. They transform electrical signals into physical actions.
   D. They store and analyze IoT data.

3. Which IoT communication protocol uses a publish/subscribe pattern?
   A. MQTT
   B. CoAP
   C. DNS-SD
   D. IPv6

4. A company wants to ensure secure data transmission for their IoT devices. What is the most important measure to implement?
   A. Using the latest firmware and encryption
   B. Allowing public access to networks
   C. Employing low-power devices without encryption
   D. Disabling network monitoring

5. What is a key pressure modern cities face?
    A. Aging population
    B. Overcrowding and pollution
    C. Vaccine shortages
    D. High healthcare costs

6. In the Business Canvas Model, what does the "Key Resources" block typically identify?
    A. The primary audience for marketing
    B. Those who can help the organization
    C. The profit model for stakeholders
    D. The infrastructure(hardware) and men required to create product

7. A custom computer manufacturer wants to reach their customers through both physical stores and online platforms. In the Business Canvas Model, where should these methods of reaching customers be listed?
    A. Value propositions
    B. Customer segments
    C. Channels
    D. Revenue streams

8. Which of the following best describes automation?
    A. A process that requires continuous human management
    B. A process that is self-driven and reduces the need for human intervention
    C. A process that eliminates all machines
    D. A process that only works in smart homes

9. What is the main difference between Artificial Intelligence (AI) and Machine Learning (ML)?
    A. AI is only used for data analysis, while ML is used for automation
    B. ML is a subset of AI that focuses on learning from data
    C. AI involves creating machines that think like humans, while ML does not
    D. AI and ML are the same thing and are interchangeable

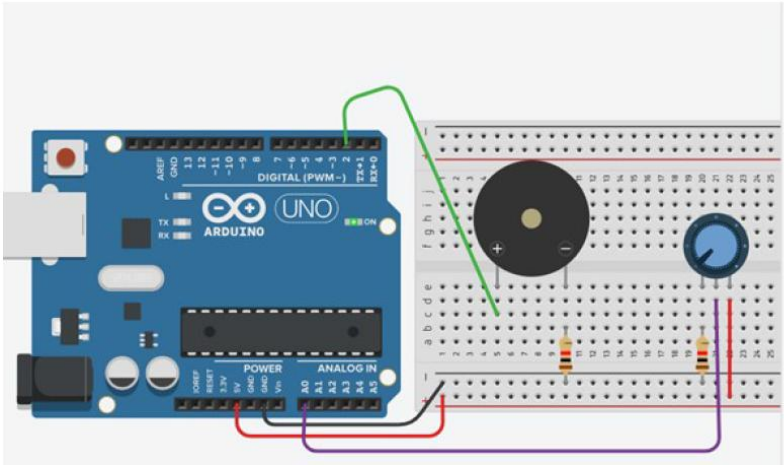
10. A small business owner is concerned about the physical security of their office. Which of the following measures would best protect the building's outside perimeter?
    A. Installing surveillance cameras around the office building
    B. Adding multi-factor authentication for employee access to the office
    C. Updating employee access control passwords regularly
    D. Disable WPA2-AES encryption for compatibility

II. **MATCHING TYPE.** (6 x 1 = 6 marks)
Match the items in column A with the items in column B. Write the number of the correct answer in the space provided.

| Column A | | Column B |
|---|---|---|
| 11. Protocol is designed to allow constrained, simple devices to join the IoT via constrained networks of low availability and low bandwidth | _____ | A. ZigBee |
| 12. A low-energy, low-power, low-data rate wireless protocol used to create personal area networks, with data requests identified by an Application Profile ID Number. | _____ | B. Value Proposition |
| 13. It's a block in the Business Model Canvas that defines the product or service the organization will provide to customers. | _____ | C. Assets Tracking |
| 14. System that tracks the hospital resources like wheelchairs, beds and equipment's | _____ | D. CoAP |
| 15. This utilizes statistical techniques to allow computers to learn from their environment and enhance performance on tasks without direct programming. | _____ | E. Educate user |
| 16. One of the security best practices for ensuring protection in an organization | _____ | F. Machine Learning |
| | _____ | G. Mining |

III. **SHORT ESSAY.** (5 marks)
Answer all questions briefly and clearly.

17. Which pin and method are used to read the input from the potentiometer in the given circuit? ( 2 marks)




```
void setup()
{
  pinMode(2, OUTPUT);
  pinMode(A0, INPUT);
}

void loop()
{
  int pvalue=analogRead(A0);
  analogWrite(2, pvalue);
}
```

18. Evaluate the any two (2) security challenges arising from the growing adoption of smart home technology. (2 marks)

**ROUGH WORKS**

Total Mark: **20** Duration: **80 minutes**

**PRACTICAL PART**

Name: _____ ID Number: _____

Section: _____ Computer Number: _____

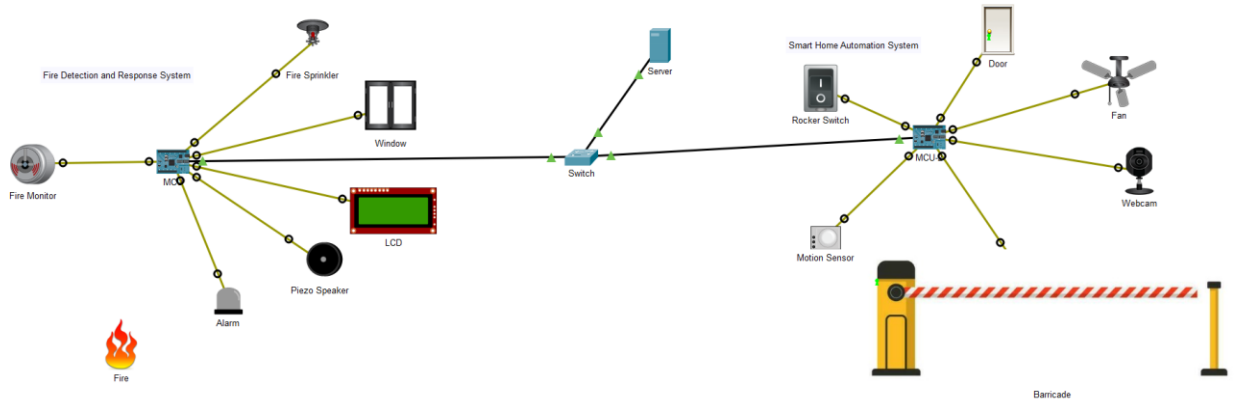
**General Instructions:**

- Create a folder in **Desktop** and name it as **ST _ID_YOURNAME_Section**
  Example: **SE_66S123_ALI_1.** SAVE ALL your files in this folder.
- The required source files are available in the folder **CSSE2204_ST** on the Desktop.
- After finishing your examination, copy your folder to the **Z:\ drive** (for exam account) and inform your invigilator.
- Please **DON'T LOG OFF** or **SHUTDOWN** your computer after the examination to avoid losing your files.

IV. **PROGRAMMING**.

Provide the required topology and program or source code of the given scenario.

19. You are tasked with developing an IoT-based smart home automation and fire detection and response system using microcontroller units (MCUs) as shown in the diagrams below. Your system must perform various automated tasks to enhance safety, security, and convenience.




1. Add the following devices to the logical workspace. (1.5 marks)
   a. IoT devices:
      Fire, Fire Monitor, MCU, Fire Sprinkler, Window, LCD, Piezo Speaker, Rocker Switch, Motion Sensor, Fan, Door, Webcam and Barricade
   b. Network device: Switch 2960
   c. End devices: Server
2. Assign a IP address 10.10.10.1 and subnet mask 255.255.255.0 to the Server. (0.5 mark)
3. Enable ethernet port on MCUs and connect the devices as shown above figure.

(0.5 mark)

4. Enable DHCP on the Server and set the following parameters to the existing serverPool. (1 mark)
   a. Starting IP address: 10.10.10.2
   b. Subnet Mask: 255.255.255.0
5. Enable DHCP on Ethernet interface of MCUs. Confirm whether the devices have a valid IP address (10.10.10.x). (0.5 mark)
6. Fire Detection and Response System (8 marks)

Develop a program to continuously monitor the fire situation

When the fire monitor detects a fire, trigger the following actions:

a. Activate the Fire Sprinkler to suppress the fire.
b. Sound the Piezo Speaker to alert the residents.
c. Display the status on the LCD screen ("FIRE").
d. Automatically open the Window to improve ventilation in case of fire.

When the fire is not detected, the following actions should be performed:

a. The fire sprinkler system should remain inactive.
b. The piezo speaker should be turned off to avoid unnecessary alarms.
c. Window should be closed to maintain security and comfort.
d. Display the status ("NO FIRE") on the LCD screen.

7. Smart Home Automation System ( 8 marks)

Develop a program perform specific actions based on input from different sensor and rocker switch to control home appliances and security features.

Fan and Door Control:

a. If the rocker switch is turned ON, the fan should be turned ON and the door should automatically open.

b. If the rocker switch is OFF, the fan should be turned OFF and the door should remain closed.

Motion Detection and Barricade System:

a. If the motion sensor detects any movement, the barricade should automatically open and the webcam should be turned ON to monitor the activity.

b. If no motion is detected, the barricade should close and the webcam should be turned OFF.

**ROUGH WORKS**